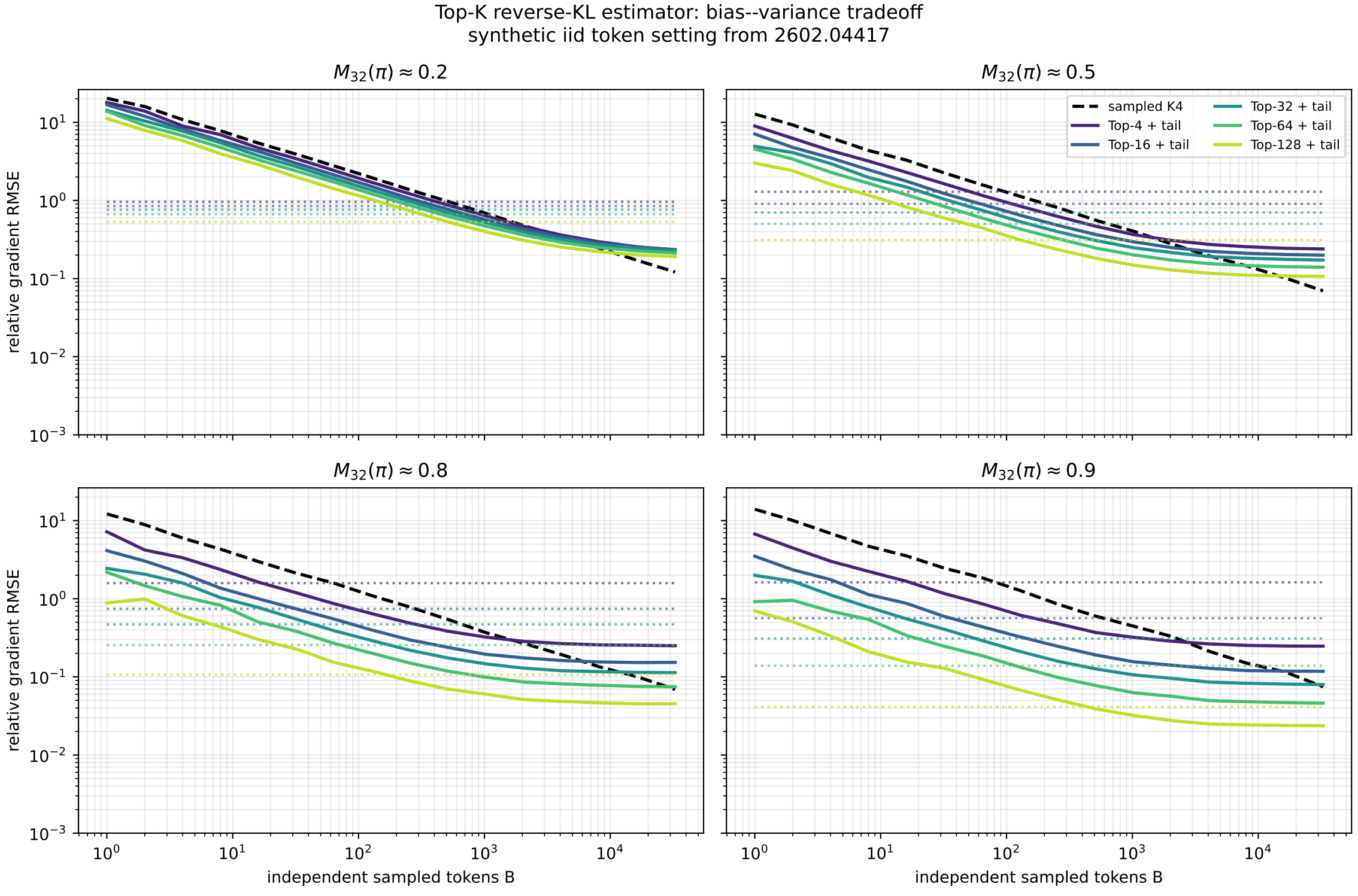
Top-K reverse-KL estimator: bias--variance tradeoff
synthetic iid token setting from 2602.04417
$M_{32}(\pi) \approx 0.2$
$M_{32}(\pi) \approx 0.5$
$M_{32}(\pi) \approx 0.8$
$M_{32}(\pi) \approx 0.9$
sampled K4
Top-4 + tail
Top-16 + tail
Top-32 + tail
Top-64 + tail
Top-128 + tail
relative gradient RMSE
independent sampled tokens B